Contents

One Guitar Band

Fig. 1: The Graphical User Interface (GUI) of One Guitar Band

Installation

At this stage, One Guitar Band does not require a formal installation procedure. Simply download the **OGB.zip** package, extract its contents, and save the resulting **OGB** folder to a location on your hard drive, for example:

C:/OGB

To launch the application, double-click **OneGuitarBand.exe**.

Important: Avoid placing the OGB folder inside **C:/Program Files** or other protected system directories, as Windows may restrict the application's ability to write files and save user data.

Optional: You can create a **shortcut** of **OneGuitarBand.exe** and save it on your Desktop so as to make it easier to access the program. Do not move **OneGuitarBand.exe** outside OGB folder as this will prevent the application from launching!

System requirements

One Guitar Band is a standalone software application developed in C++ and designed to run on desktop and laptop computers.

The current release is available for Microsoft Windows. Versions for macOS and Linux are under consideration for future releases.

To use One Guitar Band, you will need:

- A computer running Windows.
- An audio interface providing at least two analog inputs and two analog outputs.
- ASIO-compatible audio drivers (recommended for optimal performance).

While One Guitar Band may operate with a computer's built-in audio device under certain circumstances, the best user experience is achieved with a dedicated external audio interface.

Audio interface configuration

For reliable real-time performance, One Guitar Band requires specific audio settings.

Open the **Options** menu, click **Audio/MIDI settings** and configure the following parameters:

- Disable Mute Audio Input if it is enabled.
- Select ASIO as the Audio Device Type.
- Set the Sample Rate preferably to 48.000 Hz, otherwise to 44.100 Hz.
- Set the Audio Buffer Size to either:
    - 64 samples (recommended), or
    - 128 samples.

A buffer size of 64 samples provides the lowest possible latency and is therefore recommended whenever your system can support it. If your hardware cannot operate reliably at 64 samples, use 128 samples instead.

Buffer sizes below 64 samples are not supported by the application. Buffer sizes of 256 samples or higher will increase latency to a level that may become noticeable during performance and reduce the responsiveness of the system.

Once configured, these settings are stored automatically and will be recalled the next time the application is launched.

Figure 2 illustrates the recommended configuration for a Focusrite audio interface.

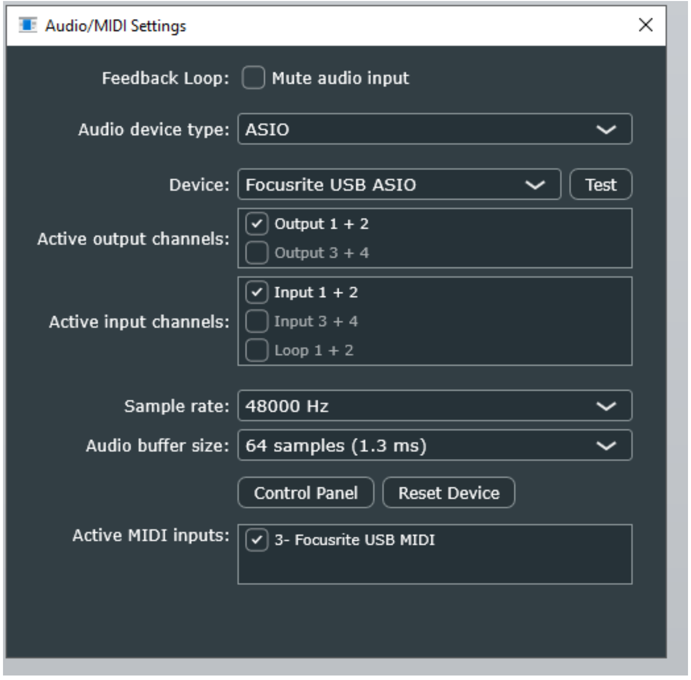

Fig. 2: Audio configuration for the case of a Focusrite soundcard

Application Overview

One Guitar Band is built around a loop-based workflow that allows musicians to construct complete arrangements by recording and combining multiple musical layers over successive loop repetitions.

The software supports:

- Four Guitar layers
- Four Modulation layers
- Four Drum layers
- Four Bass layers
- Three Cymbal voices (Hi-Hat, Ride, and Crash)

All Guitar and Modulation layers can play simultaneously. For Bass and Drums, one layer can be active at a time, as selected through the LayerIdx control.

The following sections describe the functionality of each area of the user interface shown in Figure 1.

Input Section

The Input section controls the level of the direct guitar signal routed from the audio interface inputs to the output. Separate controls are provided for the left and right input channels. Additional checkboxes allow the user to determine whether the direct guitar signal remains audible while recording Modulation, Bass, Cymbals, or Drums. In most cases, keeping these options enabled is recommended, as it helps maintain a clean monitoring environment during recording.

Guitar Section

The Guitar section controls the playback level of each recorded Guitar layer.

For every layer, the user can adjust:

- Volume
- Stereo Position (Pan)
- Delay Send Level

The Pan controls determine the placement of each layer within the stereo field, while the Send controls determine how much of the signal is routed to the internal delay processor.

InputCh: This controller lets the user specify which audio channel is fed to each guitar Layer. Set the selector to "L" to record the left channel, "R" to record the right channel or "L+R" to perform a stereo audio recording.

PostProcessing: Pressing the "p" button opens a new window where the user can apply processing to the already recorded guitar layers, or to guitar layers that have been loaded. Specifically, that user can

- **Pitch shift**: transpose the pitch of a guitar layer from one octave down to one octave up.
- **Time stretch**: time stretch a guitar layer so as to match the loop duration. This is particularly useful when loading guitar recordings with durations that don't exactly match the current loop duration.
- **Sample rate conversion**: use this functionality when a guitar layer that has been loaded does not match the sampling rate of the project.

In that same window, the user can find **Save** and **Load** functionalities to save a guitar layer that has been just recorded or to load previously recorded guitar layers or audio files.

Clear: Click this button to completely erase a specific guitar layer (i.e., the one selected through **LayerIdx** control).

Export: Exports the selected guitar layer as a .WAV file. All exported audio files can be found at the following path ".../OGB/data/users/exports/".

Modulation Section

The Modulation section allows users to transform previously recorded Guitar layers by altering their rhythmic and tonal characteristics. Up to four Modulation layers can be created, corresponding directly to the four available Guitar layers. Each Modulation layer operates exclusively on its matching Guitar layer:

- Modulation Layer 1 accepts as input Guitar Layer 1
- Modulation Layer 2 accepts as input Guitar Layer 2
- And so on

If no audio has been recorded in the corresponding Guitar layer, the Modulation layer will produce no output.

Four modulation engines are available and each one has it's own unique character and relates to different parameter dials, as shown in the Talbe below:

Modulation type	**Useful parameter dials**	**Affected by MIDI Control Change (CC)**
Decay	Decay, NonlinDecay, Send	Decay
Frequency	CentralFreq, Threshold, Decay, NonlinDecay, Send	CentralFreq
Spectral	Decay, NonlinDecay, Alpha, Send	Decay

Chirp	Decay, NonlinDecay, Threshold, Alpha, Send	Alpha

For example, when **mode = Frequency**, the sound is affected by the position of CentralFreq, but that is not the case when **mode= Decay**. Knob **Alpha** is intended for use with **mode = Spectral** and **mode = Chirp** and will not affect the sound for other modes.

One Guitar Band accepts MIDI Control Change (CC) messages for continuously adjusting control parameters associated with the Modulation mechanisms. For example, a MIDI expression or volume pedal can be used to continuously modify parameter values in real time. The type of parameter affected by CC messages depends on the modulation type, as can be seen at the third column of the Table above.

The rhythmic behavior of the generated modulation depends on the selected Trigger mode:

Trigger = Input

The modulation follows the input signal produced by gestures performed by the player on the guitar.

Trigger = Cymbal/input

The modulation follows the rhythmic events previously recorded in the Cymbals layer.

Trigger = T, T/2, T/3, T/4

The modulation follows subdivisions of the metronome, producing rhythmically synchronized modulation patterns

Trigger = Drums

The modulation follows the rhythmic events previously recorded in the Drums section

Parameters such as Decay, NonLinDecay, Frequency, and Threshold only influence the sound during the recording stage. Once a Modulation layer has been recorded, changes to these controls will not affect the existing recording. To modify the resulting sound, the layer must be recorded again using the desired settings.

Clear: Click this button to completely erase a specific modulation layer (i.e., the one selected through LayerIdx).

Export: Exports the selected modulation layer as a .WAV file.

Bass Section

The Bass engine generates bass lines directly from guitar performance gestures.

Two independent bass voices are available: **Vibration** and **Impulse**

These voices may be used independently or simultaneously, allowing a wide range of bass textures and performance styles.

Up to four Bass layers can be recorded. However, only one Bass layer can be active at any given time, as determined by the **LayerIdx** selector.

Pitch Tracking: The Bass engine follows the pitch information performed on the three lowest guitar strings, covering the range from E1 to A2. It assumes that the guitar is tuned at 440.0 Hz.

Vibration Voice: The Vibration voice continuously follows the actual vibration of the guitar strings and responds in real time to the player's performance. This voice is ideal for smooth bass lines that constitute the low frequency end of the composition.

Impulse Voice: The Impulse voice is driven by string attacks rather than sustained vibrations. When the **Trigger** is set to **pick**, the system detects pick or finger attacks and uses them to generate bass events. Unlike the Vibration voice, these events are processed after recording has been completed. Alternatively, the Impulse voice can be synchronized to the metronome using rhythmic subdivisions such as: T, T/2, T/3 or T/4. The Impulse voice generally produces a tighter and more staccato sound compared to the Vibration voice.

PostProcessing: When using the Impulse voice with **Trigger** = **pick**, post processing mechanisms can be enabled to correct minor timing inaccuracies and improve rhythmic consistency.

- When setting **PostProc** to **classifyOnsets**, the recorded guitar sound will be re-analyzed by the offline algorithm with the goal to provide a more accurate estimation of the picking time instants.
- When setting **PostProc** to **quantize**, the recorded guitar sound will be re-analyzed by the offline algorithm with the goal to provide a more accurate estimation of the picking time instants and to quantize the bass sequence along the time grid.

Clear: Click this button to completely erase a specific bass layer.

Copy/Paste: Use these buttons to copy a bassline from one layer to another.

Export: Exports the selected bass LayerIdx as a .WAV file. Different .WAV files will be produced for Impulse and Vibration voices.

Cymbals Section

The Cymbals section provides access to three different cymbal voices:

- Hi-Hat
- Ride
- Crash

In addition to their musical role within an arrangement, cymbals also function as a valuable timing reference during recording. By providing a rhythmic guide, the Cymbals section helps musicians maintain consistent timing while recording additional layers.

Trigger Options

Depending on the Trigger setting, cymbal events can be generated in several ways.

- <u>Metronome-Based Triggering</u> :The cymbal voices can follow rhythmic subdivisions of the metronome, including T, T/2, T/3, T/4 and 3T. This approach effectively turns the Cymbals section into a customizable metronome.

- Input-Based Triggering: When Trigger is set to Input, cymbal events follow the percussive gestures recorded by the player.
- Drum-Based Triggering: cymbal events can also be derived from recorded Drum patterns, allowing the creation of coherent drum-and-cymbal arrangements with minimal additional programming.

Clear: Click this button to completely erase the cymbal element that corresponds to the ID selector.

Exports: Exports a specific cymbal as a .WAV file.

Drums Section

The Drums section provides control over the three fundamental drum components used by One Guitar Band:

- Kick Body
- Kick Attack
- Snare

Up to four Drum layers can be recorded and stored. As with Bass layers, only one Drum layer can be active at a time, selected via the **LayerIdx** control. Drum patterns are generated by performing specific percussive gestures on the guitar. Detailed instructions for these gestures are provided later in the chapter *Drum Performance Gestures.*

Drum Models: Two different generic machine-learning models are available: Acoustic and Electric. Select the model that matches the type of guitar being used.

- Choose **Acoustic** for acoustic or electro-acoustic guitars.
- Choose **Electric** for electric guitars.

For creating and using a Custom model, read the Section *Drum Performance Gestures.*

Post-Processing and Quantization: The **PostProc** selector provides optional post-processing tools.

- **off**: no correction is applied to the played sequence
- **correct**: the captured guitar signal is re-analyzed by a more accurate offline algorithm. Potentially, this can correct classification errors (e.g. kicks identified as snare or the opposite) made by the real-time algorithm
- **quantize:** the captured guitar signal is re-analyzed by the offline algorithm and moreover, drum hits are quantized along the time axis. In most of the cases, this improves the rhythmic precision of recorded drum performances.

Fusion: When **fuse** is ticked, drum gestures from more than one drum layers are fused together to produce a single sequence for triggering the drum synthesis engine. For example, the user can record only the kicks at Layer 1 and the snares at Layer 2. If Fuse is ticked for Layer 2, the outcome will be a sequence containing both kicks and snares. This setting is useful for dance music styles where simultaneous kick and snare events are quite common. The characteristics of the produced sound will depend on the settings of the current drum layer. Note that fusion sums the velocities from Layer 1 up to the selected drum Layer, while Layer with a higher number do not participate in the fusion process. For example, when fuse is ticked for Layer 3, then Layers 1, 2 and 3 participate in the fusion process but Layer 4 doesn’t.

Use the Section buttons to enable or disable drum fusion in combination with the active Drum layer.

Mapping of gestures to drums: The **Mapping** selector determines the mapping between playing gestures and drum sounds.

- Input -> k+s : employs the chosen drum recognition model (i.e. acoustic, electric or custom) to transform each hit into a kick or a snare sound
- Input -> k: playing gestures are translated to kick sound only
- Input -> s: playing gestures are translated to snare sound only
- None: use this setting as an alternative way for muting drums

Important note: the desired mapping choice must be selected prior to recording the control sequence (the playing gestures). Changing the mapping after the gestures have been recorded will not affect the produced drum sequence.

Clear: Click this button to completely erase a specific drum layer.

Copy/Paste: Use these buttons to copy a drum sequence from one layer to another.

Export: Exports the selected drum LayerIdx as a .WAV file. Different .WAV files will be produced for kick-body, kick-attack and snare.

Output Section

The Output section contains the controls used for monitoring and managing the final audio signal.

Master Output Level: dedicated control allows adjustment of the overall output volume of the application.

Level Meter: The integrated level meter provides visual monitoring of both input and output signal levels. When the application starts, the meter displays the input signal level by default. Pressing the **Input/Output** selector changes the meter source:

- Input mode displays incoming audio levels.
- Output mode displays the final output level.

This makes it easy to monitor signal strength throughout the recording and playback process.

Delay Controls: The Output section also contains the controls associated with the internal delay processor. The delay effect can be applied to Guitar, Bass and Modulation layers, allowing users to add spatial depth and rhythmic complexity to their arrangements.

Song Sections: The buttons located in the lower-right corner of the interface allow transitions between different song sections. Song Sections are associated to different arrangement during performance and are discussed in detail in a following chapter.

Input signal routing

For One Guitar Band to operate correctly, a clean (dry) guitar signal must be connected to the first (**Left**) or second (**Right**) input channel of the audio interface.

For best results, avoid inserting effects processors in the clean signal path. The more heavily processed the incoming signal becomes, the more difficult it is for the software's analysis and gesture-recognition algorithms to perform reliably.

However, if desired, users may optionally route a processed (wet) version of the guitar signal to the other remaining input channel.

This allows the software to receive a clean signal for analysis while simultaneously monitoring a processed guitar tone through external effects units, amp simulators, or pedalboards.

To achieve this configuration, the guitar signal must be split into two parallel paths:

- Dry signal → Audio Interface Input 1 (Left) or Input 2 (Right)
- Wet signal → Audio Interface Input 2 (Right) or Input 1 (Left)

Signal splitting can be accomplished using a splitter cable, a DI box or a dedicated splitter device.

This routing configuration is illustrated in Figure 3.

By default, the software assumes that the dry guitar signal is routed to Input 1.

If for some reason you prefer the software to consider Input 2 as the dry input signal, click the **Customize** button at the lower right corner of the GUI and select **R** as the **Dry Guitar Channel**. The software will remember this choice every time you open it and therefore it will not be necessary to apply this setting every time.

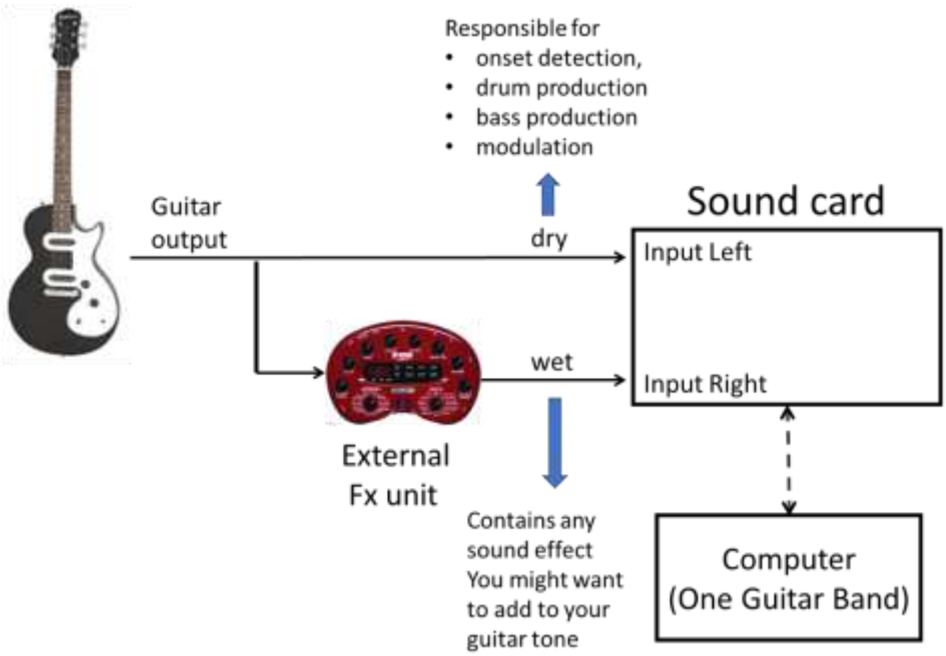

Fig. 3: Routing configuration

Figure 4: Percussive gestures for drum production with an acoustic guitar (left) and an electric guitar (right)

Drum performance gestures

One of the defining features of One Guitar Band is its ability to generate drum sounds directly from percussive gestures performed on the guitar. The program comes with two generic gesture recognition models, one for **acoustic** and one for **electric** guitars. The generic models are compatible to both sampling rates of 48000 and 44100 Hz. Once a generic model is chosen, the user should perform the gestures in a specific way so as to obtain accurate recognitions results. Except from the generic models, we offer the ability for the user to create custom models, based on gestures of his/her preference. Both the generic model, based on the recommended gestures, and the custom model, based on user- defined gestures are covered in what follows.

Drum production with the generic models.

Drum production using the acoustic or electric generic model requires gestures to be produced in a specific way. Although they are very easy to perform, a short familiarization period is recommended in order to achieve the best recognition accuracy. The demonstration videos available on the One Guitar Band YouTube channel provide valuable examples of the recommended playing techniques.

Classic, Acoustic and Electro-Acoustic guitar

To generate a **kick drum** sound, strike the guitar body using the side of your thumb. The objective is to produce a sound with strong low-frequency content and significant body resonance (see Fig. 4 left). Lightly resting part of the palm on the guitar body helps towards dampening high frequency vibrations, decreasing the risk of kick gestures being recognized by the model as snares.

To generate a **snare** sound, strike the soundboard using the tip of your index or middle finger near the edge of the guitar top. It is important to allow some amount of **fingernail** contact. The goal is to produce Increased high-frequency content and if possible, a clearly audible wooden character. Use a wearable pluck or some other solid object if you think that the system misses snare hits or confuses them too often with kicks. Impact with hard solid objects increase high-frequency energy and help the recognition algorithm distinguish the snare gesture from the kick gesture.

Electric Guitar

Before attempting drum gestures on an electric guitar, it is recommended to set the pickup selector to the bridge pickup position.

For most guitars this corresponds to:

- The far-right position on Fender-style instruments
- The lowest position on Epiphone-style instruments

This ensures that the bridge pickup, which is closest to the bridge, is responsible for capturing the signal.

Generate the **kick** sound by striking the lowest string with your thumb as shown in Figure 4. Occasionally, touching the second string is acceptable and generally does not affect recognition performance.

Generate the **snare** sound by striking the upper frets using your index or middle finger. This action excites one or more of the thinner strings and creates the transient characteristics required by the recognition system.

Avoid excessively muting the strings with the palm of your picking hand, as this may suppress the signal and reduce recognition accuracy. At the same time, try to prevent prolonged ringing by lightly damping the strings with your fretting hand (e.g. the left hand in Fig. 4) after each strike.

The ideal gesture produces:

- A strong transient
- A clearly detectable signal
- Minimal sustain

Relevant video for drum production with an electric guitar: https://youtu.be/CZ_Vj-RB094

Creating a custom model for drum production

Custom models for drum production can be easily created by following a simple training phase. Press the "Customize" button at the lower right corner. This will open a secondary window with additional buttons required for training the system and saving or recalling a custom model.

Press the "Custom kicks" button and repeat the gesture intended for triggering the kick sound 10 to 15 times. Once done, press the "Custom kicks" button again to terminate the recording process.

Press the "Custom snares" button and repeat the gesture intended for triggering the snare sound 10 to 15 times. Once done, press the "Custom snares" button again to terminate the recording process.

Press the "Save" button in the same secondary screen to save the model in the form of an XML file. Once done, the model will be available in location

…/OGB/data/users/G2D/customModels/

so that the user can recall it at any other time instant.

Important notes:

- It is a good practice to cover a wide dynamic range when recording percussive gestures, i.e., from the weakest to the strongest hits that may occur during performance.
- Avoid picking up gestures that are acoustically similar, since this will also make it hard for the recognition model to discriminate gestures intended for kick from those intended for snare production.
- A custom model generated at 48000 Hz sampling rate might not work so well at 44100 Hz and vice versa. Generate different models at 48000 and 44100 Hz if you intend to work at both sampling rates.
- Read and follow the instructions on threshold adjustment provided in the next Section since the yellow threshold setting might affect the outcome for the training phase.

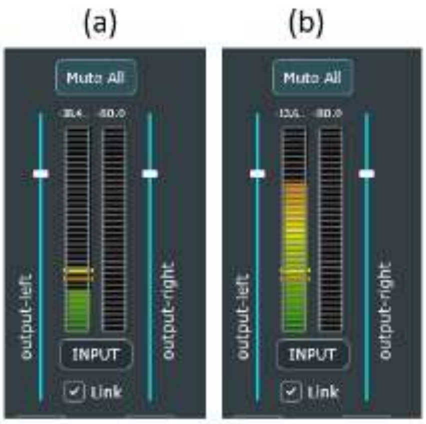

Fig. 5: Threshold adjustment is important for Bass, Cymbals και Drums production.

Threshold adjustment for Bass, Cymbals και Drums

Proper threshold adjustment is essential for reliable operation of the Bass, Cymbals, and Drums engines. Two threshold controls are provided:

- **Orange Threshold:** Bass Section
- **Yellow Threshold:** Cymbals and Drums Sections

The Cymbals and Drums engines share the same threshold setting.

How Threshold Detection Works: When generating Bass or percussion sounds, the software expects the incoming signal level to exceed the corresponding threshold value. Signals below the threshold are treated as background noise and do not trigger sound generation. Figure 5 illustrates both cases of incoming signal strength below and above the threshold values.

Recommended Adjustment Procedure: Use the level meter located on the right side of the graphical interface by setting to display **input** sound level. When performing intentional Bass or Drum gestures, signal peaks should clearly exceed the threshold level (Fig. 5 right). When the guitar is idle, the signal should remain below the threshold (Fig. 5 left). A correctly adjusted threshold prevents missed triggers and reduces false triggers.

Saving and Loading Projects

One Guitar Band supports both preset storage and arrangement storage.

Presets: Most of user-interface settings such as instrument knob settings, channel routing, trigger states and operation modes can be stored as presets in the form of XML files. When the application starts, it automatically loads the preset named “defaultSettings”.

To create your own presets:

- Press **Save** or **Save As**
- Enter a preset name
- Recall presets later using **Load**

Saving and recalling individual instruments: in addition to presets, the software can store all recorded musical layers for Guitar, Modulation, Bass, Cymbals and Drums. This is possible through the **Load** and **Save** buttons available below each instrument section.

This allows complete arrangements to be recalled and further developed at a later time.

The **Load All** function loads all instruments simultaneously. For this feature to work correctly all instrument files must share the same filename and the filename must match the preset name.

For Example, if:

- Preset = "song1"
- Drums = "song1"
- Guitar = "song1"

Then pressing **Load All** and choosing preset “song1” restores the complete arrangement.

If the Guitar file was instead saved as "song2", the Guitar layer will remain empty when loading "song1".

Storage Formats: Bass, Cymbals, and Drums are stored in a symbolic representation similar to MIDI. As a result, they automatically adapt to tempo changes. Guitar and Modulation layers are currently stored as WAV files and therefore do not automatically follow tempo changes.

Exporting to DAW

Every instrument layer can be exported as an individual WAV file, by pressing the **Export** button found in each instrument section. For Guitar, Modulation, Bass and Drums, the exported WAV file will correspond to the layer specified in **LayerIdx** controller. For Cymbals, the exported WAV file will correspond to the element specified in **ID** controller. This makes it easy to transfer projects from One Guitar Band to a Digital Audio Workstation (DAW) for further recording, mixing and editing towards the final music production.

Export Specifications: Exported WAV files are saved at the project's sample rate (48000 or 44100 Hz) and 16-bit resolution. The duration of the exported file matches the duration of the loop.

BPM adjustment: The BPM of each project is displayed on the upper left of the GUI. This information can be exploited so at to set the DAW project to have the same BPM as your One Guitar Band project.

Important Note: For Bass and Drum layers, the application must continue playback until at least one complete loop cycle has elapsed after pressing **Export**. Exported files are automatically named according to the current preset name and can be found in:

.../OGB/data/users/exports

Future releases are expected to include MIDI export capabilities for Bass, Cymbals, and Drums.

Song sections

The song Sections panel is located at the lower right part of the user-interface and provides a powerful way to move instantly between different arrangement during performance. They are particularly useful in live situations where instruments need to enter or leave the arrangement at specific moments.

What Is Stored in a song Section?

Sections store:

- **Mute** states for all instruments and layers
- **LayerIdx** values (for Drums and Bass)
- **Trigger** states for each one of the three cymbal elements
- **Fusion** states for Drum layers

Sections do not store:

- Instrument parameter values
- Mode
- Loop duration

Those settings remain part of the preset itself.

Note that, unlike the case of Cymbals, the **trigger** state for Drums, Bass and Modulation is fixed with the **LayerIdx** and does change with **Section** index.

Saving and Loading Sections

The Section panel provides five available sections slots: S1, S2, S3, S4 and S5.

Pressing **Save** stores the current state into the selected slot.

Pressing **Load** recalls the stored state.

Immediate vs Quantized Switching: Settings associated to Sections S1–S4 are applied immediately. Section 5 behaves differently: The requested change is queued and occurs automatically at the beginning of the next loop cycle.

Similar to Presets, **Sections are saved as XML files** and can be found inside preset folder at "OGB/data/users/presets/". The Section files are distinguished from the corresponding preset files by the fact that their name ends with "_trans.xml".

MIDI setup

One Guitar Band optionally accepts MIDI input, allowing the user to control several of the app's functionalities through MIDI commands.

Press MIDI SETUP, near the upper left corner of the GUI, to customize the mapping between MIDI commands and One Guitar Band functionalities. This will open a secondary window displaying all the functionalities that can be triggered through MIDI commands. Next to each functionality, type the number that corresponds to the desired MIDI command. Typically, MIDI devices output note numbers from 0 to 127, therefore integer values should be within this range.

To finalize the setup, press the SAVE button. The mapping will be saved and it will be automatically recalled every time that the app is opened.

In case that you intend to use an expression pedal or some other interface to control Modulation parameters (see Section Modulation) through MIDI CC messages, you can use the slider at the bottom of the MIDI SETUP window to customize the mapping between control input values and modulation parameter values. The slider allows simultaneous adjustment of a minimum and maximum threshold and the two thresholds are saved together with all other settings in the corresponding preset XML files.

Important notes:

- The program cannot accept blank values for the note numbers. It will automatically assign MIDI note numbers to functionalities that the user has left blank. Therefore, it is a good advice to setup all fields, including those that you do not intend to use.
- Ideally, all functionalities should be assigned different MIDI note numbers to avoid confusion. This is important especially for commands associated to the recording process. However, you should feel free to assign the same MIDI number to a recording functionality and to a Section change command. For example, if **recordGuitar1** and **section3** are assigned the same MIDI note number, then both events can be triggered with the same MIDI command without conflicting with one another.
- Functionalities named **recordGuitar1, recordGuitar2**, **recordGuitar3**, or **recordGuitar4** allow the user to initiate the recording process for the desired guitar layer. On the other hand, **recordGuitar** will initiate the recording process for the guitar layer that matches the value of **LayerIdx** displayed in

the GUI. The same also holds for all other recording functionalities, i.e., **recordBass**, **recordModulation** and **recordDrums** will initiate the recording process for the layerIdx value that is displayed on the corresponding instrument section in the GUI.

Recording Modes

Initiating a recording from the GUI is very easy; press GUITAR to record a guitar layer, MODULATION to record a modulation layer, BASS to record a bass layer, CYMBALS to record a cymbal layer and DRUMS to record a DRUMS layer. These are the so-called INSTRUMENT buttons. Due to their importance, each INSTRUMENT button is linked to a specific MIDI note number, so that the user can interact with them not only from the GUI but also through the MIDI device (if available). Each one of the five INTRUMENT buttons is responsible

- For initiating the recording for a particular instrument,
- for arming the instrument so that the recording can begin immediately after an onset is detected,
- for terminating the recording process for a particular instrument, or
- for terminating the recording process for the previous instrument and arming the recording process for the next instrument.

There are two choices for simultaneously initiating the recording and playback process (i.e. the **Stop** button has been pressed):

- **Loop Starts with onset:** recording is automatically initiated the moment that the program detects an acoustic onset (strong attack + signal level above a threshold).
- **Loop Starts with MIDI:** the recording is initiated the moment that the user presses the corresponding instrument button on the GUI or on the MIDI controller.

When the app is in playback mode, acoustic onset detection has actually no effect on the recording process.

The **Mode** selector, located at the middle left of the GUI, is one of the most important controllers of One Guitar Band, offering four different ways for controlling the recording process.

- **Metronome, Keep Last Take:** As the name implies, this mode relies on a predefined loop duration. The app will keep recording until an INSTRUMENT button is pressed once more, which will signal termination of the recording process. The application will preserve the last completed take. Example: assume that you are happy with the GUITAR take that you recorded at the fourth repetition of the loop. Press the GUITAR button at any moment during the fifth loop cycle. The last completed take (i.e. the one captured during the fourth repetition of the loop) will be preserved while all other takes will be erased and completely ignored. **Why This Workflow Is Important:** The combination of **Metronome, Keep Last Take** along with **Loop Starts with Onset** allows recording to begin and end with minimal interaction with the user interface. As a result, the musician can remain focused on performance rather than operating the application.
- **Metronome, overwrite:** this mode also relies on a predefined loop duration. In this case however, the end of the recording process will be defined by the instant that the user presses the corresponding INSTRUMENT button on the GUI or on the MIDI controller. Overwriting implies that

the recorded audio stream is constantly overwritten. Example: Assume that you are recording a bass sequence and that you press the BASS button on the GUI (or on the MIDI controller) at the fourth repetition of the loop. The final BASS sequence will be comprised by two parts; the first part will contain gestures played on the fourth repetition of the loop while the second part will contain gestures performed during the third repetition of the loop.

- **No metronome, overwrite**: Similar to the previous mode, the recording process is terminated at the instant that the corresponding INSTRUMENT button is pressed on the GUI (or on the MIDI controller). The difference with the previous mode is that the loop duration is not predefined. It will be automatically defined the moment that the user terminates the recording process for the first time after it was initiated. This approach resembles the typical behavior found in standalone hardware loopers. For obvious reasons, a footswitch MIDI controller is a strong prerequisite when using this mode.
- **Metronome, just 1 take**: this mode also relies on a predefined loop duration. The recording process will begin at *t*=0 and will continue until one loop cycle is completed. No need for the user to terminate the recording process as this will occur automatically. Unlike with the other modes, pressing an INSTRUMENT button while another instrument is being recorded will not terminate the recording process. Instead, the program will wait until the cycle is completed and will initiate the recording for the new component immediately at the end of the loop.

Getting Started

When One Guitar Band is launched, the preset **defaultSettings** is loaded automatically.

The following examples are designed to help new users become familiar with the software's workflow and understand some of its most important features.

Scenario #1 – Creating Your First Loop

Step 1: Start Playback

Press the **Start** button located in the lower-right corner of the interface. You will hear a hi-hat pattern that functions as the application's built-in metronome. The timing of this metronome is determined by the current loop settings.

Step 2: Adjust Tempo

You can modify the tempo by adjusting **Loop Duration** located on the left side of the interface. This value represents the duration of the loop in seconds. The loop duration can be any number between 0.4s and 48 s. You can also change the rhythmic subdivision by modifying: **NumTimeDivisions.** Increasing this value produces more metronome hits per loop, while decreasing it produces fewer. When you are satisfied with the tempo, press **Stop**.

Understanding the Recording Mode:

By default, the **Mode** selector is set to: **Metronome, Keep Last Take.** This mode uses a fixed loop duration and automatically stores the most recently completed take. Each time you record a layer, the previous recording is replaced by the latest complete performance.

Directly below the Mode selector, you will find the option: **Loop Starts with Onset**. Assume that playback is stopped and no audio is currently playing. Now press the **GUITAR** button to arm a Guitar layer for recording. You will notice that recording does not begin immediately. Instead, the status message displays: **Waiting for Onset.** The software is now waiting for the first note or gesture. As soon as you play the guitar, recording and playback begin automatically. This allows the musician to start recording naturally without having to coordinate button presses with performance timing.

Step 3: Record Your First Guitar Layer

Play until at least one complete loop cycle has elapsed. Then press the **GUITAR** button again to stop recording. Congratulations — you have just created your first loop!

Step 5: Build an Arrangement

Experiment with: Additional Guitar layers, add a Bass layer, a Drum layer, or a Modulation layer.

Scenario #2 – Exploring a Pre-Built Arrangement

Step 1: Load the Demonstration Project

Press **Load All** and select "**defaultSettings".** This loads the demonstration arrangement supplied with the software.

Step 2: Start Playback

Press **Start**. You will hear a complete arrangement consisting of previously recorded instruments and layers.

This project is intended to demonstrate the capabilities of the software and provide a starting point for experimentation.

Step 3: Explore the Controls

While playback is running, experiment with:

- Volume controls
- Pan controls
- Delay settings
- Bass sounds
- Drum sounds
- Cymbal settings

Observe how each parameter affects the overall mix.

Step 4: Perform song Section changes

Press, S2 or S3 and compare the resulting arrangements.

Notice how instruments and layers enter or leave the mix, changing the musical structure without interrupting playback.

This demonstrates how the Section panel can be used to create verse/chorus changes, arrangement variations, and live-performance dynamics.

Trial Version Limitations

The trial version of One Guitar Band includes a small number of restrictions.

Audio Interruption: To distinguish the trial version from the full version, audio output is automatically muted periodically during operation.

Export Functions: Exporting and saving functionalities are disabled in the trial release.

Unlocking the Full Version

To obtain a fully functional version, press the “Copy Machine ID” button, located at the bottom of the GUI, to copy your Machine ID to the clipboard. Send us this information by email at onegtrband@gmail.com

In less than 24h you will receive by email a link for downloading a personalized version of **OneGuitarBand.exe** that operates without the limitations of the trial release on your computer.

Support and Feedback

One Guitar Band is an actively evolving project, and user feedback plays a crucial role in its development.

If you require assistance with:

- Installation
- Configuration
- Performance optimization
- Workflow questions
- MIDI integration

please contact us by email at onegtrband@gmail.com

We are committed to helping users achieve the best possible experience with the software.

Personalized Optimization

In certain cases, we may be able to optimize aspects of the software specifically for your playing style.

Examples include:

- Gesture-recognition improvements
- Performance configuration suggestions
- Customized machine-learning models

Online Demonstrations

Depending on demand and availability, we may also organize video calls to:

- Demonstrate features
- Explain advanced workflows
- Help troubleshoot issues
- Gather user feedback

Additional Learning Resources

A more comprehensive user manual is currently under development. In the meantime, many aspects of the software can be learned through the videos available on our official channels.

YouTube:
www.youtube.com/@OneGuitarBand

Instagram:
https://www.instagram.com/OneGuitarBand/

Frequently Asked Questions

Is One Guitar Band Available for Free?

One Guitar Band is currently in an evaluation and testing phase and has not yet been released as a commercial product.

Offering the software free of charge to a limited number of users allows us to:

- Collect valuable user feedback
- Identify opportunities for improvement
- Refine the user experience
- Validate new features in real-world use

We also hope that satisfied users will help introduce One Guitar Band to a wider audience and contribute to the growth of the project.

Which Types of Guitars Are Supported?

One Guitar Band works with most conventional guitars.

The recommended options are:

- Electric guitars
- Electro-acoustic guitars with built-in pickups

For acoustic guitars without built-in electronics, an external pickup is recommended.

Piezo pickups mounted directly on the guitar body generally provide the best results.

However, acceptable performance may also be achieved using microphones in certain situations.

Do I Need Headphones?

Headphones are strongly recommended but not mandatory.

Some of the software's algorithms may be affected by loud external sounds, including sounds generated by the application itself. This issue is most noticeable when using acoustic guitars. Headphones greatly reduce the possibility of acoustic feedback or false triggering. That said, many users successfully operate One Guitar Band through amplifiers or loudspeakers.

Performance depends on:

- Playback volume
- Guitar type
- Pickup type
- Acoustic environment

With electric guitars, acoustic leakage is typically negligible, allowing significantly higher monitoring levels.

Do I Need a MIDI Controller?

No. All core features of One Guitar Band are fully accessible through the graphical user interface.

However, a MIDI controller can significantly enhance the experience, especially during live performance.

A MIDI controller allows hands-free operation of:

- Recording functions
- Layer switching
- Section changes
- Performance controls

If you already own a MIDI controller and would like to integrate it with One Guitar Band, please follow the instructions in Section **MIDI Setup**.

Hosting of third party plugins

One Guitar Band does not currently provide the ability to host external plugins or virtual instruments. Nevertheless, any instrument layer produced within One Guitar Band can be exported as a WAV file and consequently, imported into the Digital Audio Workstation of your preference for further processing. We are currently working on functionalities that will allow the user to export Bass, Cymbals and Drums also in MIDI format. This will make it possible to use gestures produced with One Guitar Band to trigger third party virtual instrument.

Is One Guitar Band based on AI?

The answer to this question depends on how you define AI. One Guitar Band employs machine learning technology for the drum and bass production mechanisms. However, machine learning is limited to real-time or offline classification purposes. Our program does not employ generative AI technologies, in the sense that the program does not generate music from zero and the end user is responsible for every single note or beat that is produced by the software.

Final Remarks

One Guitar Band was created with a simple goal:

to enable a single guitarist to build complete musical arrangements in real time using only a guitar and a computer.

Whether you are sketching ideas, practicing, composing, recording demos, or performing live, we hope the software inspires new creative possibilities and helps you explore new ways of making music.

Thank you for supporting the project and being part of the One Guitar Band community.